

# VEHICLE RESALE PRICE PREDICTION

## Regularized Polynomial Regression Pipeline

*A Machine Learning Project Report*

### 1. Project Background & Objective

The goal of this project is to construct an end-to-end machine learning pipeline to predict used car prices based on vehicle specifications and history. By moving from raw, noisy data to an optimized, regularized polynomial model, the pipeline demonstrates how systematic cleaning, leakage-free feature engineering, and regularization directly impact model generalization.

### 2. Deep Dive: Data Preprocessing & Cleaning

#### Raw Data Profile

- Initial Shape: 4,340 rows × 8 feature columns.
- Features: name, year, selling\_price, km\_driven, fuel, seller\_type, transmission, owner.

#### Cleaning Operations Executed

**Deduplication:** Dropped 763 exact duplicate entries, leaving 3,577 unique observations.

**Missing Value Check:** Verified 0 null values across all features.

**Anomaly Mitigation:** Detected a data entry error at row 1312 — a 2014 Mahindra Quanto listed with km\_driven = 1 under “Second Owner,” flagging it as an unreliable outlier during inspection.

**Outlier Capping (Winsorization):** Extreme feature distributions were capped at the 1st/99th percentile thresholds to prevent cost-function distortion during gradient descent / normal equation solving:

- selling\_price: capped from ₹8,900,000 down to ₹2,675,000.
- km\_driven: capped from 806,599 km down to 223,158 km.
- year: lower bound capped from 1992 up to 2000.

### 3. Deep Dive: Feature Engineering & Data Integrity

To convert raw textual attributes into model-ready features without leaking target or test distribution knowledge:

#### 1. Ordinal Encoding (owner)

Categorical string levels were mapped into sequential integer ranks reflecting valuation impact: First Owner → 1, Second Owner → 2, Third Owner → 3, Fourth & Above Owner → 4, Test Drive Car → 5.

#### 2. Dimensionality Reduction (name → brand)

The raw name column contained 1,491 granular sub-model variants — extremely high cardinality that, if one-hot encoded directly, creates a sparse feature space large enough for the model to effectively memorize individual listings rather than learn generalizable pricing patterns.

Solution: the leading manufacturer word was parsed from each entry (e.g., “Maruti 800 AC” → “Maruti”), collapsing the name column into 29 distinct brand categories before one-hot encoding alongside transmission, fuel, and seller\_type.

3. Data Leakage Mitigation in Feature Scaling

**Problem:** Scaling the whole dataset before splitting introduces test-set variance and mean into the training parameters.

**Solution:** The pipeline splits data first into training and test sets (80/20). StandardScaler was fit strictly on the training data and subsequently applied to transform the test set.

4. Technical Progression & Benchmark Performance

Three modeling iterations were benchmarked, moving from an unregularized linear baseline to a fully tuned, scaled polynomial Ridge model:

| Iteration     | Architecture & Strategy                     | Key Hyperparameter | Train R <sup>2</sup> | Test R <sup>2</sup> | Assessment                                                                                           |
|---------------|---------------------------------------------|--------------------|----------------------|---------------------|------------------------------------------------------------------------------------------------------|
| Baseline (v1) | Standard Linear Regression                  | N/A                | —                    | 0.6641              | Baseline model; no polynomial interaction terms.                                                     |
| Iter 1 (v2)   | Degree-2 Polynomial + Ridge (unscaled)      | $\alpha = 5.0$     | 0.741                | 0.646               | High variance. Unscaled km_driven <sup>2</sup> (10 <sup>9</sup> magnitude) dominated the L2 penalty. |
| Final (v3)    | Scaled Degree-2 Poly + Ridge (GridSearchCV) | $\alpha = 0.5$     | 0.732                | 0.698               | Optimal generalization. +5.2 pts R <sup>2</sup> over v2; train-test gap narrowed sharply.            |

Hyperparameter Tuning: GridSearchCV over Ridge  $\alpha$

Rather than picking a regularization strength by hand, the final iteration used GridSearchCV with 5-fold cross-validation across nine candidate values of  $\alpha$ , evaluated on the scaled, degree-2 polynomial feature set (numeric features standardized before expansion). The sweep below shows mean cross-validated R<sup>2</sup> at each candidate  $\alpha$ :

| Alpha ( $\alpha$ ) | Mean CV R <sup>2</sup> (5-fold) | Std Dev |
|--------------------|---------------------------------|---------|
| 0.001              | 0.7108                          | 0.0414  |
| 0.01               | 0.7113                          | 0.0410  |
| 0.1                | 0.7125                          | 0.0397  |
| 0.5 (best)         | 0.7126                          | 0.0383  |
| 1.0                | 0.7114                          | 0.0361  |
| 5.0                | 0.6946                          | 0.0235  |

|       |        |        |
|-------|--------|--------|
| 10.0  | 0.6780 | 0.0177 |
| 50.0  | 0.6216 | 0.0092 |
| 100.0 | 0.5873 | 0.0076 |

Cross-validated performance peaks in the  $\alpha = 0.01\text{--}0.5$  range and degrades sharply beyond  $\alpha = 5$ , confirming that only light regularization is needed once features are properly scaled — heavier penalties simply shrink useful coefficients toward zero and underfit.

Final Selected Model

| Metric                                    | Value       |
|-------------------------------------------|-------------|
| Best Regularization Strength ( $\alpha$ ) | 0.5         |
| Train $R^2$                               | 0.732       |
| Test $R^2$                                | 0.698       |
| Test MAE                                  | ₹152,741.03 |
| Test RMSE                                 | ₹226,292.33 |

5. Key Engineering Learnings & Future Enhancements

**Impact of Scale on Regularization:** Ridge's L2 penalty ( $\alpha\sum\beta_i^2$ ) fails when feature scales differ significantly. Scaling numerical values prior to polynomial expansion was the key driver behind the jump from a 0.646 to a 0.698 test  $R^2$ .

**Impact of Cardinality on Generalization:** Reducing name to brand before one-hot encoding was equally critical — without it, the model has enough sparse dummy columns to fit individual listings almost exactly (train  $R^2$  above 0.97) while test performance stagnates, a textbook case of high-cardinality overfitting.

**Model Ceiling:** Linear architectures max out around  $\sim 0.70$   $R^2$  due to non-linear resale depreciation curves (e.g., luxury car vs. budget car depreciation rates).

**Next Logical Step:** Benchmark tree-based ensemble algorithms (Random Forest Regressor or XGBoost) to capture complex non-linear feature interactions naturally.

*Prepared as part of an academic machine learning project.*